

Rawan Shehayib

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EDUCATION

University of Notre Dame | Notre Dame, IN

May 2028

Bachelor of Engineering

GPA: 3.75

Major: Computer Engineering | Minor: Energy Studies

Full Scholarship Recipient | Transformational Leaders Program | Galvin Science and Engineering Scholar

University College Dublin - Engineering Study Abroad | Dublin, Ireland

Summer 2025

RESEARCH EXPERIENCE

IrishSat | Notre Dame, IN

August 2024 – Present

Lead Computing and Controls Engineer

- Oversee the computing and controls function of a student-built satellite, setting technical direction and aligning 15+ cross-functional engineers around a shared roadmap
- Coordinated with external partners (NearSpace, SpaceX) ahead of Notre Dame's first satellite launch, ensuring all software technical requirements were met
- Built a Python-based sensor management system for a Raspberry Pi satellite simulator, reducing debugging time by 40% and accelerating the team's development cycle
- Designed computer vision algorithms for autonomous horizon detection, validated against 80+ images, enabling decision-making without ground-station intervention

Trinity College Dublin | Dublin, Ireland / Remote

May 2026 – Present

Research Assistant, Machine Learning for Audio Signal Processing

- Developed a fine-tuning and multimodal retrieval framework to improve detection reliability and enable archive-wide pattern search to detect environmental events across years
- Resolved data problems by standardizing 5,445 human annotations across inconsistently formatted files into a unified 16-class taxonomy, aiming at a consistent pre-trained model evaluation
- Presenting findings at an upcoming poster session with a live user-interface model, translating technical results for a non-technical audience

Kyutech University | Fukuoka, Japan / Remote

May 2026 – July 2026

Underwater Robotics Intern

- Built a native iOS (SwiftUI) control application connecting to ROS2 via WebSocket, enabling real-time teleoperation with live video feed and connection signal monitoring.
- Solved multiple network configuration failures across all three environments, taking the pipeline from simulation to live hardware control
- Collaborated with a four-member international team across Japan and the U.S. to validate the control system on physical USV hardware

MetoyerLab in Human-Computer Interaction | Notre Dame, IN

August 2025 – May 2026

Undergraduate Research Assistant

- Collaborated with a 4-person team to build an AI-powered nutrition platform using the OpenAI API, solving a 3-way tradeoff between cost, nutrition, and dietary goals using optimization algorithms
- Supported research where results were published at CHI'26 through engaging directly with 55 low-income households evaluated across an 8-week study, collecting and processing dietary data to power personalized recommendations

LEADERSHIP AND ACTIVITIES

CS4Good | Notre Dame, IN

November 2025 – Present

Team Lead

- Led a 5-member team developing a math-learning platform for local community centers, defining the project roadmap, assigning deliverables, and managing the collaborative development workflow

Notre Dame Mathematics Department | Notre Dame, IN

July 2025 – August 2025

Calculus Tutor

- Tutored 10+ incoming Engineering students in Calculus I & II, tailoring explanations to individual learning gaps

TECHNICAL AND LANGUAGE SKILLS

Technical: Python, C, C++, MATLAB, Java, Git, Linux, PCB design, SolidWorks, SystemVerilog, FPGAs

Language: English (Native), Arabic (Native), French (Intermediate)